

Chapter 6b:

Application of Newton's Laws (2)

+

Chapter 13:

Gravitation

Learning Objectives

6.3 Centripetal Force

6.4 Drag Force and Terminal Speed

13.1 Newton's Law of Universal Gravitation

13.2 Gravitation near the Earth's surface

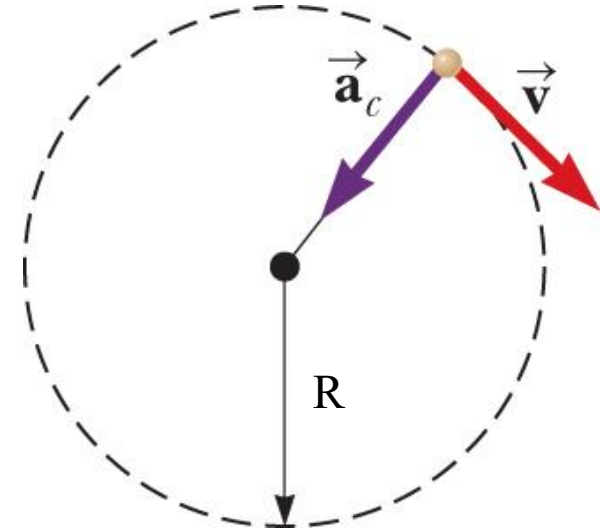
13.4 Satellite Orbit

Useful Videos

- Centripetal Force:
<https://www.youtube.com/watch?v=KvCezk9DJfk>
- Banking of Roads:
<https://www.youtube.com/watch?v=eGZWVwcaq0U>
- The One Design Change That Made NASCAR Races Faster:
<https://www.youtube.com/watch?v=AYvdS73abgI>
- Terminal Velocity:
<https://www.youtube.com/watch?v=p0IZsfzDS4s>
- Newtons Law of Universal Gravitation:
<https://www.youtube.com/watch?v=rHckbQtrP70>
- Professor Dave Explains – Universal Gravitation:
<https://www.youtube.com/watch?v=kxkFaBG6a-A>
- How does Satellite Television work?
https://www.youtube.com/watch?v=r0r4P1UAv_g

Circular Motion (reminder...)

- Where there's an acceleration there's a force!
- When going in a circle $a = a_c = \frac{v^2}{R}$
- $\mathbf{F}_{\text{net}} = ma = ma_c = m \frac{v^2}{R}$
- $\mathbf{F}_{\text{net}}$ = centripetal force



Centripetal Force

Definition: a force that makes a body follow a curved path

• **Variable:** $\vec{F}_c$

Unit: Newtons (N)

• **Equation:** $\mathbf{F}_c = \mathbf{F}_{\text{net}} = m\mathbf{a}_c = m \frac{v^2}{R}$

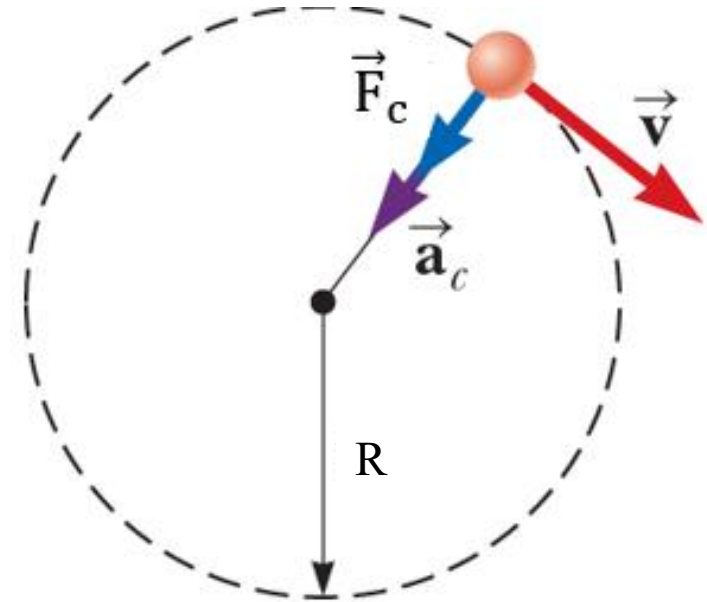
• m: mass (kg)

• R: radius (m)

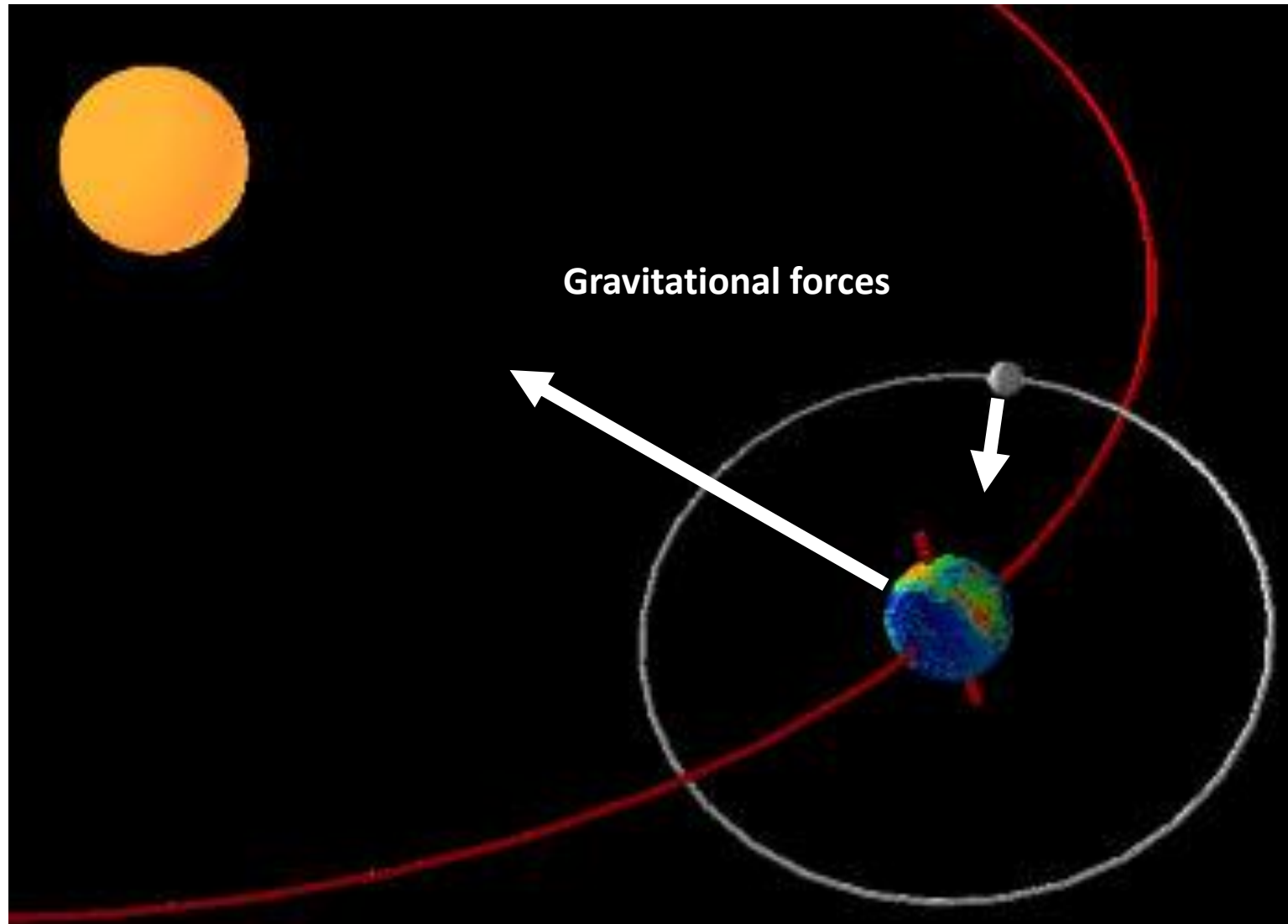
• v: **tangential velocity** (m/s)

• $\mathbf{a}_c$: **centripetal acceleration** (m/s²)

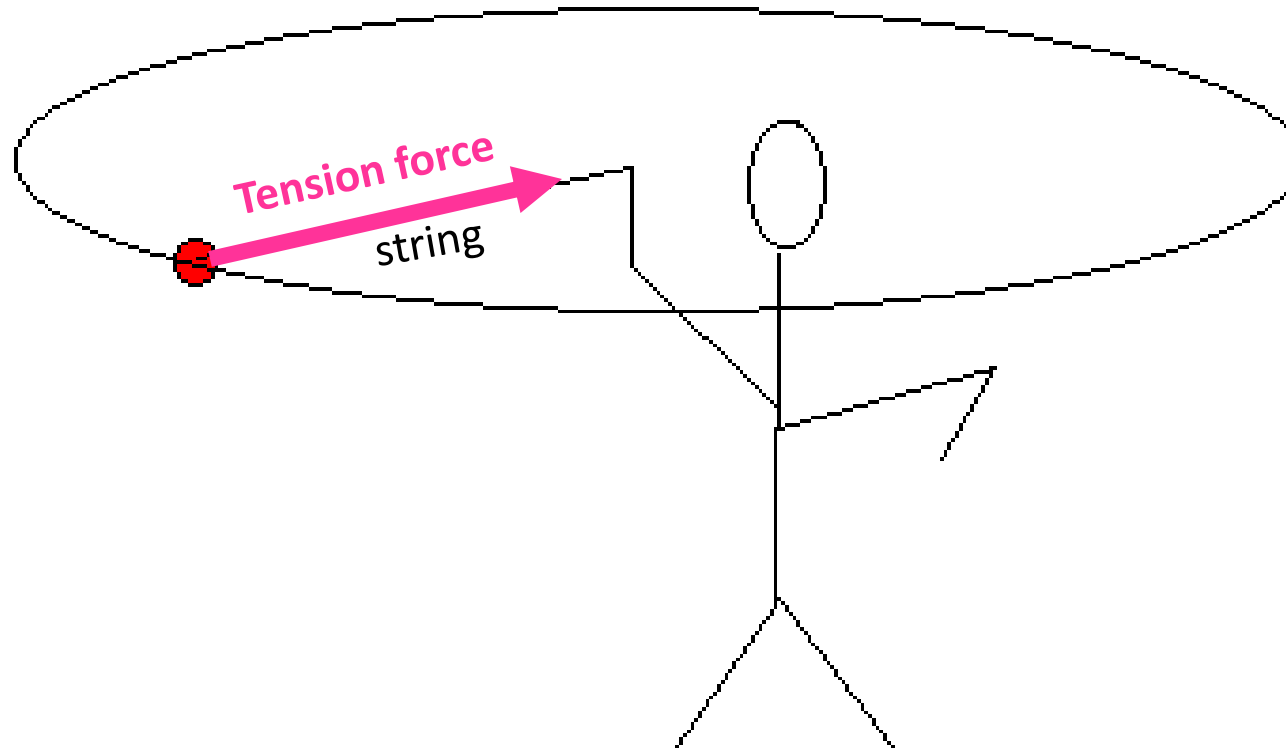
Direction: Pointed towards the center



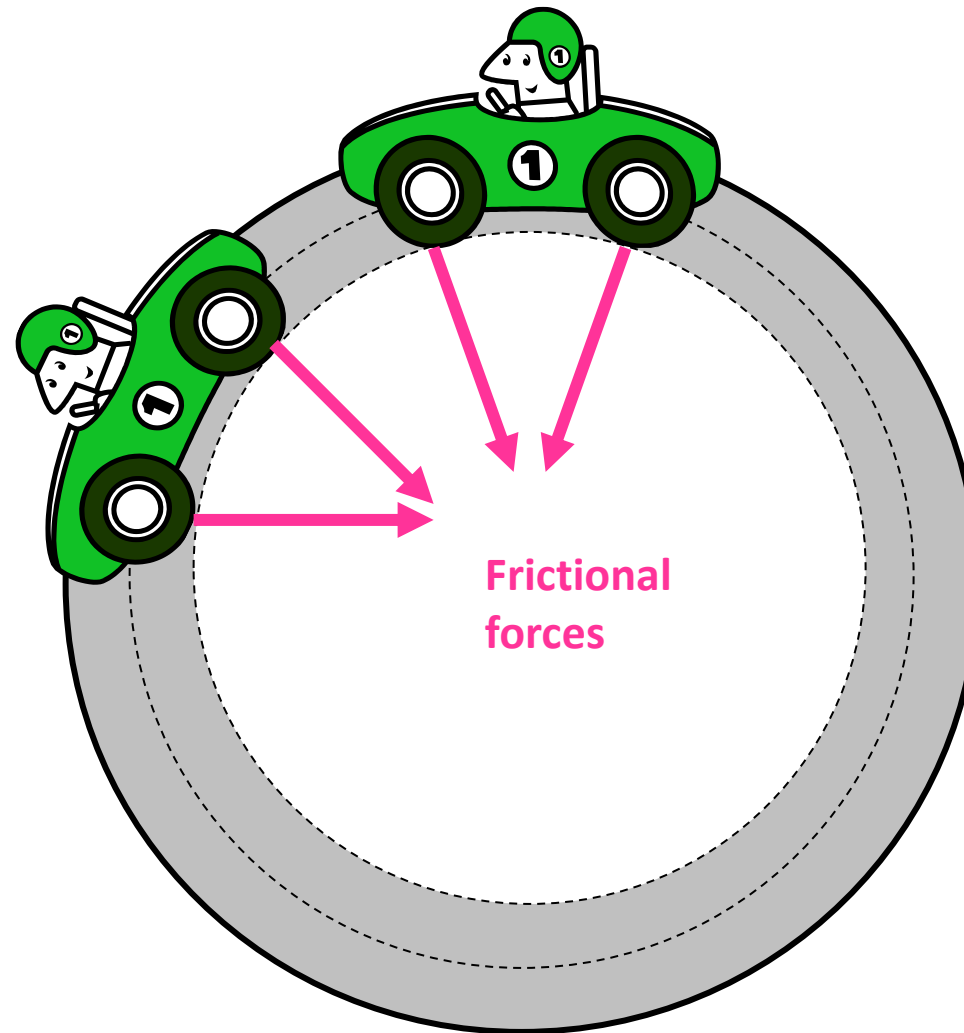
What's the **force** contributes for a centripetal acceleration here?



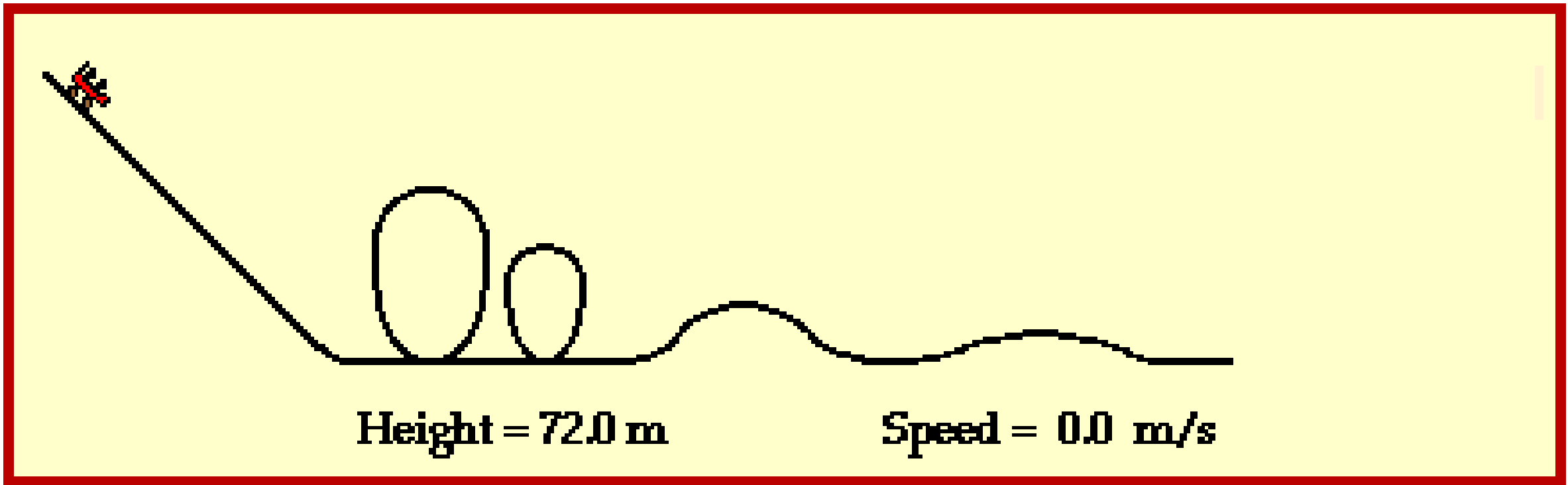
What's the **force** contributes for a centripetal acceleration here?



What's the **force** contributes for a centripetal acceleration here?

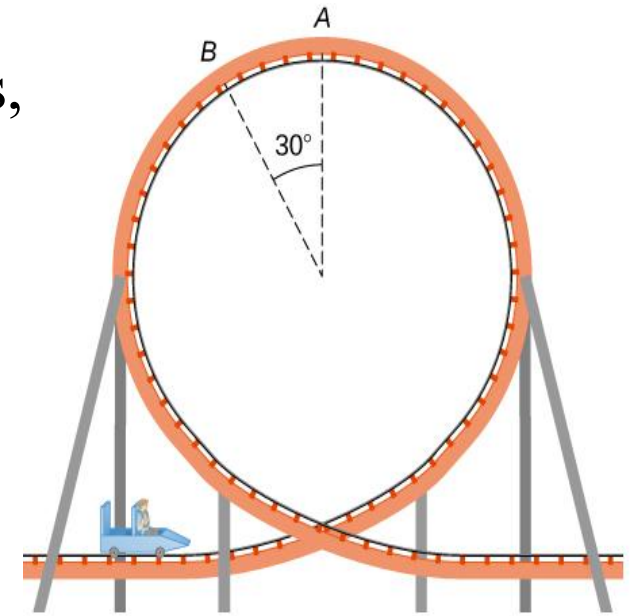


Roller coaster



Problem Solving 1 (Textbook#73)

A child of mass 40.0 kg is in a roller coaster car that travels in a loop of radius 7.00 m . At point A the speed of the car is 10.0 m/s , and at point B, the speed is 10.5 m/s . Assume the child is not holding on and does not wear a seat belt.



(a) What is the **force** of the car seat on the child at point A?

[179 N]

(b) What is the **force** of the car seat on the child at point B?

[290 N]

(c) What is the **minimum speed** required to keep the child in his seat at point A?

[8.28 m/s]

Problem Solving 1 (Textbook#73)

a) At point 'A' both forces $\vec{F}_g$ (gravitational force) and $\vec{N}$ (normal force) is directed towards the center.

$$F_c = ma_c$$

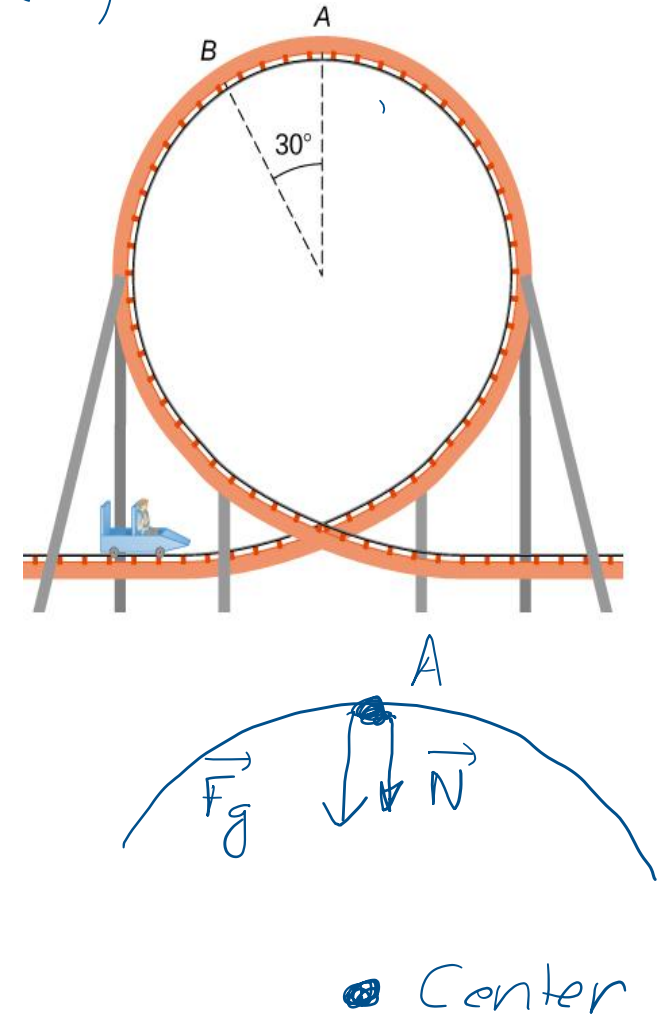
$$\vec{F}_g + \vec{N} = m \frac{v^2}{R}$$

forces pointing to the center are positive.

$$\begin{aligned}\vec{N} &= m \frac{v^2}{R} - \vec{F}_g \\ &= m \frac{v^2}{R} - mg\end{aligned}$$

$$\vec{N} = 179 \text{ N}$$

$$\begin{aligned}m &= 40.0 \text{ kg} \\ R &= 7.00 \text{ m} \\ v &= 10.0 \text{ m/s}\end{aligned}$$



Problem Solving 1 (Textbook#73)

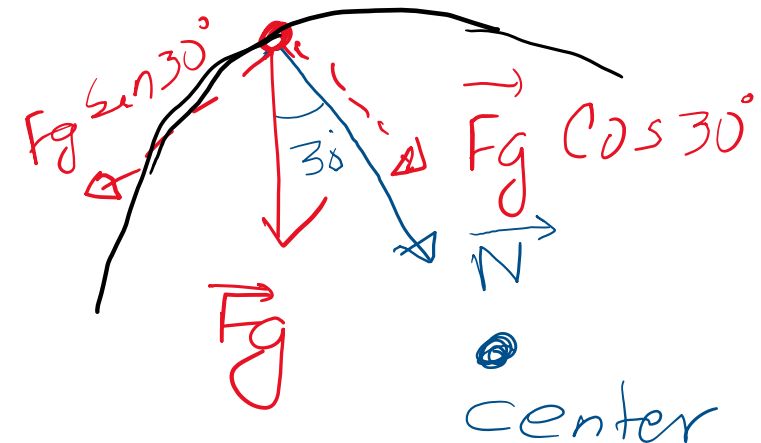
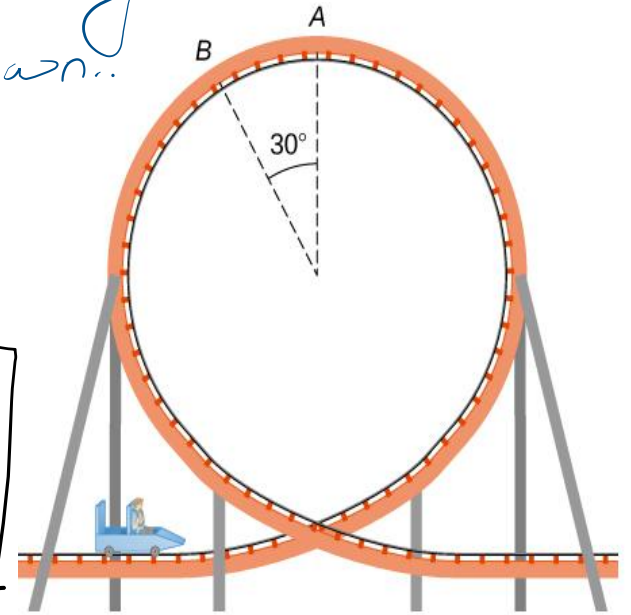
b) At point 'B', $\vec{N}$ (normal force) is pointing to the center but $\vec{F}_g$ is pointing down. The component of $\vec{F}_g$ towards the center of loop is $\vec{F}_g \cos 30^\circ$.

$$F_c = m a_c$$
$$\vec{N} + F_g \cos 30^\circ = m \frac{v^2}{R}$$

$$\vec{N} = \frac{mv^2}{R} - F_g \cos 30^\circ$$

$$\vec{N} = 290 \text{ N}$$

$$\begin{aligned} m &= 40.0 \text{ kg} \\ R &= 7.00 \text{ m} \\ v &= 10.5 \text{ m/s} \end{aligned}$$

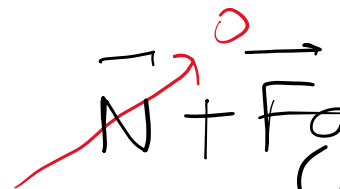


Problem Solving 1 (Textbook#73)

Q) To find minimum velocity at point 'A' we have to assume Normal force is ZERO.

Note: If the car's velocity is less than minimum velocity, it will lose contact with the track.

$$F_c = m a_c$$



A diagram of a car at the top of a loop. Two arrows point downwards from the car, labeled $\vec{N}$ and $\vec{F}_g$. A red arrow points from the text $\vec{N} + \vec{F}_g = \frac{mv^2}{R}$ to the $\vec{N}$ arrow.

$$\vec{N} + \vec{F}_g = \frac{mv^2}{R}$$

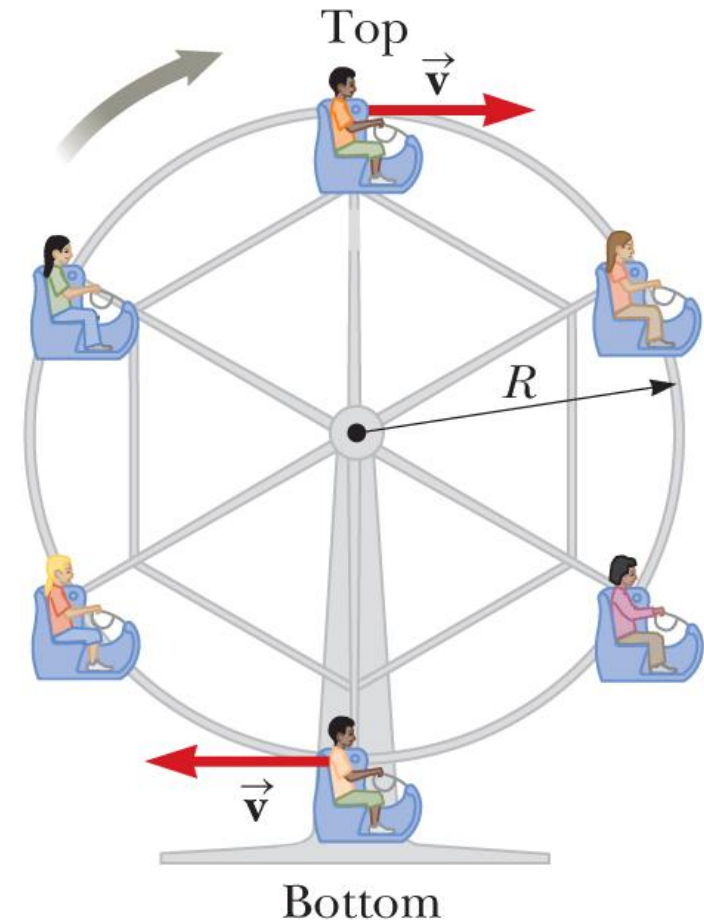
$$\cancel{mg} = \frac{\cancel{m}v^2}{R}$$

$$\Rightarrow v^2 = Rg$$
$$v = \sqrt{Rg} = \dots \text{ m/s}$$

Example 1 Riding the Ferris Wheel

A child of mass m rides on a Ferris wheel as shown in the figure. The child moves in a vertical circle of radius 10.0 m at a constant speed of 3.00 m/s .

(A) Determine the **force** exerted by the seat on the child at the **bottom** of the ride. Express your answer in terms of the weight of the child, mg .



Example 1 Riding the Ferris Wheel

(The force that is directed to the center)

$$F_c = ma_c$$

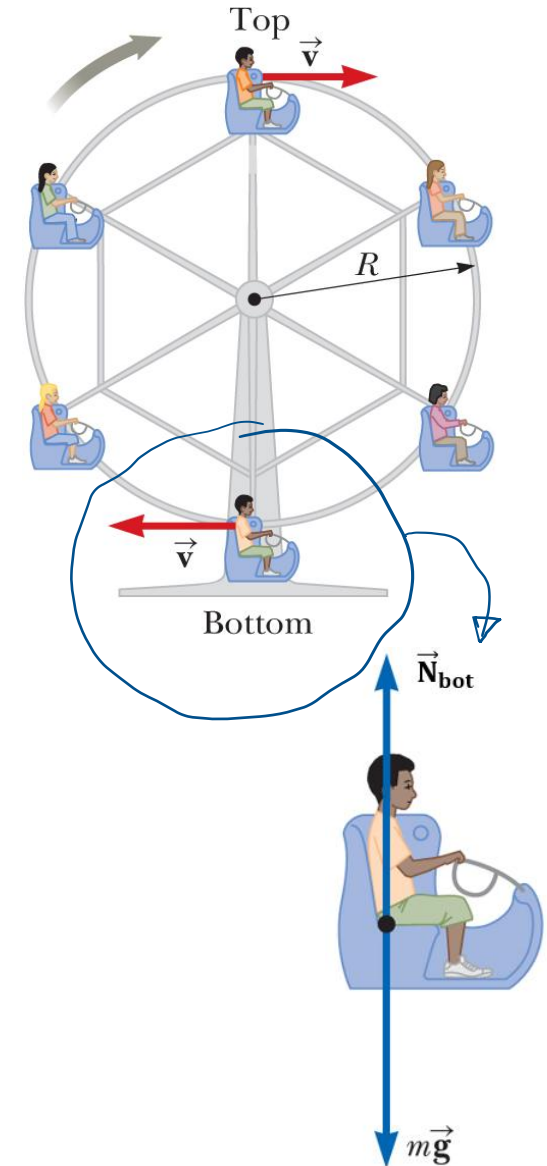
$$N_{\text{bot}} - mg = m \frac{v^2}{r}$$

$$N_{\text{bot}} = mg + m \frac{v^2}{r} = mg \left(1 + \frac{v^2}{rg} \right)$$

$$N_{\text{bot}} = mg \left(1 + \frac{(3.00 \text{ m/s})^2}{(10.0 \text{ m})(9.80 \text{ m/s}^2)} \right)$$

$$N_{\text{bot}} = \boxed{1.09 mg}$$

$$\vec{N} > m\vec{g}$$



Example 1 Riding the Ferris Wheel

(B) Determine the **force** exerted by the seat on the child at the **top** of the ride.

(The force that is directed to the center)

$$F_c = ma_c$$

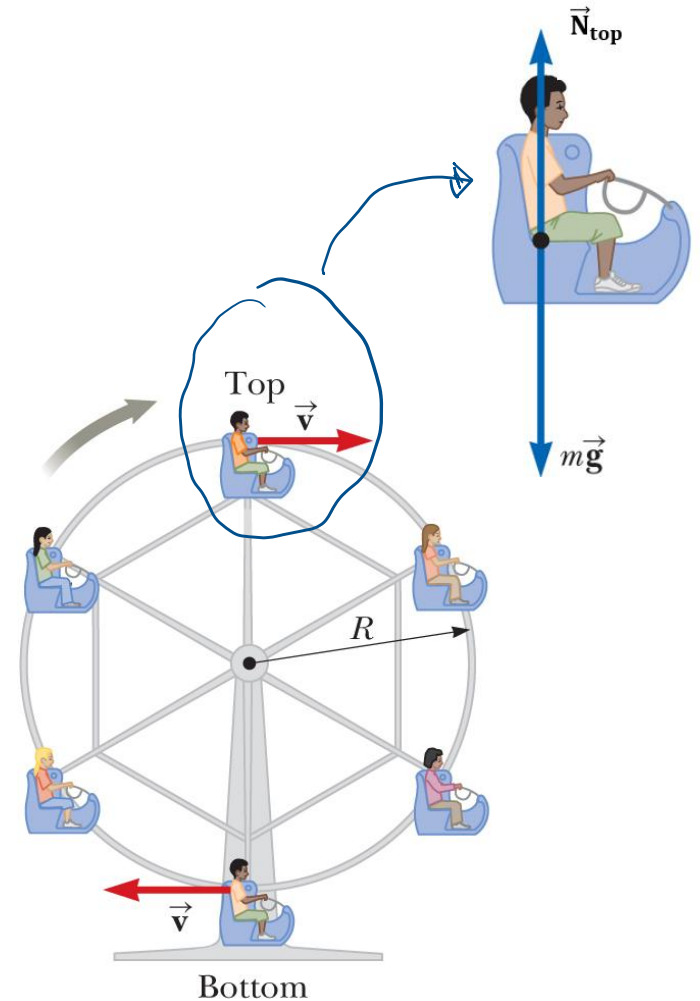
$$mg - N_{\text{top}} = m \frac{v^2}{r}$$

$$N_{\text{top}} = mg - m \frac{v^2}{r} = mg \left(1 - \frac{v^2}{rg} \right)$$

$$N_{\text{top}} = mg \left(1 - \frac{(3.00 \text{ m/s})^2}{(10.0 \text{ m})(9.80 \text{ m/s}^2)} \right)$$

$$N_{\text{top}} = \boxed{0.907 mg}$$

$N < mg$



Flat/Unbanked curve and Banked curve

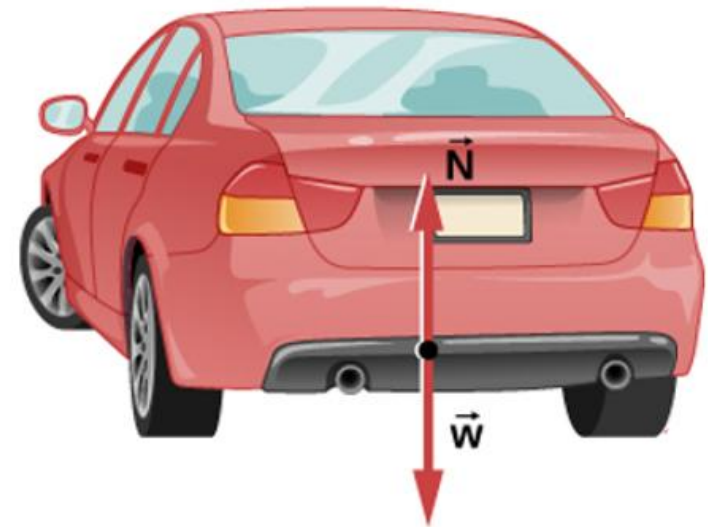


ABCD card

Suppose a car is making a left turn on a flat (unbanked) curve. Calculate the **centripetal force** exerted on a 900 kg car as it negotiates a curve with a radius of 500 m at a speed of 25.0 m/s.

- a. 112.5 N
- b. 1125 N
- c. 11.25 N
- d. 1.125 N

$$F_c = m a_c$$
$$= m \frac{v^2}{R}$$



Maximum velocity for a car on the Flat Curve

(The force that is directed to the center)

$$F_C = ma_c$$

$$f_s = \frac{mv^2}{R}$$

$$f_{s,\max} = \frac{mv_{\max}^2}{R}$$

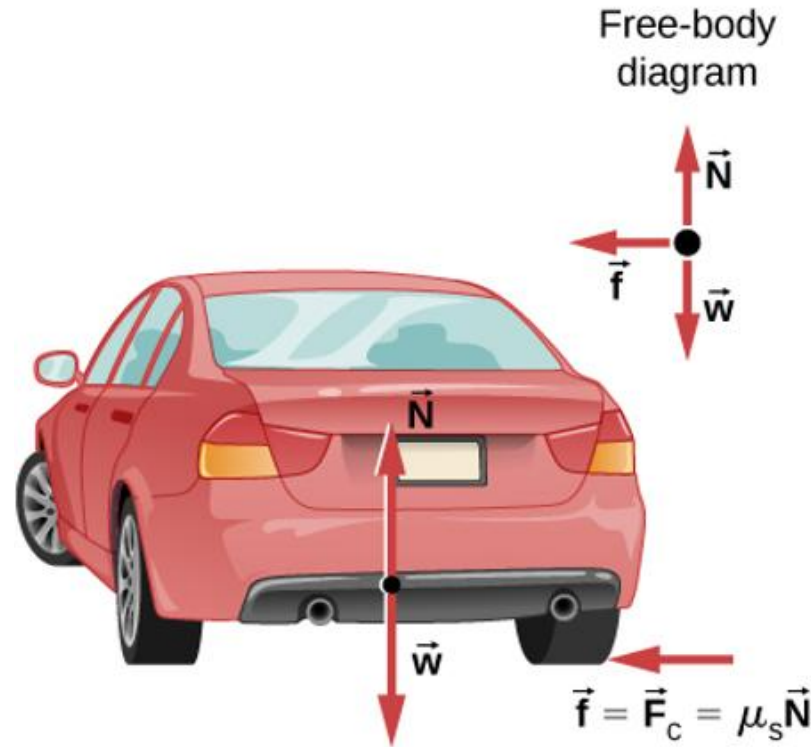
$$\mu_s N = \frac{mv_{\max}^2}{R} \dots \dots (1)$$

(No motion along y-axis)

$$\sum F_y = 0$$

$$N - mg = 0$$

$$N = mg \dots \dots (2)$$



From equation (1) and (2)

$$\mu_s mg = \frac{mv_{\max}^2}{R}$$

$$\mu_s mg R = mv_{\max}^2$$

$$\mu_s g R = v_{\max}^2 \dots \dots (3)$$

$$v_{\max} = \sqrt{\mu_s g R}$$

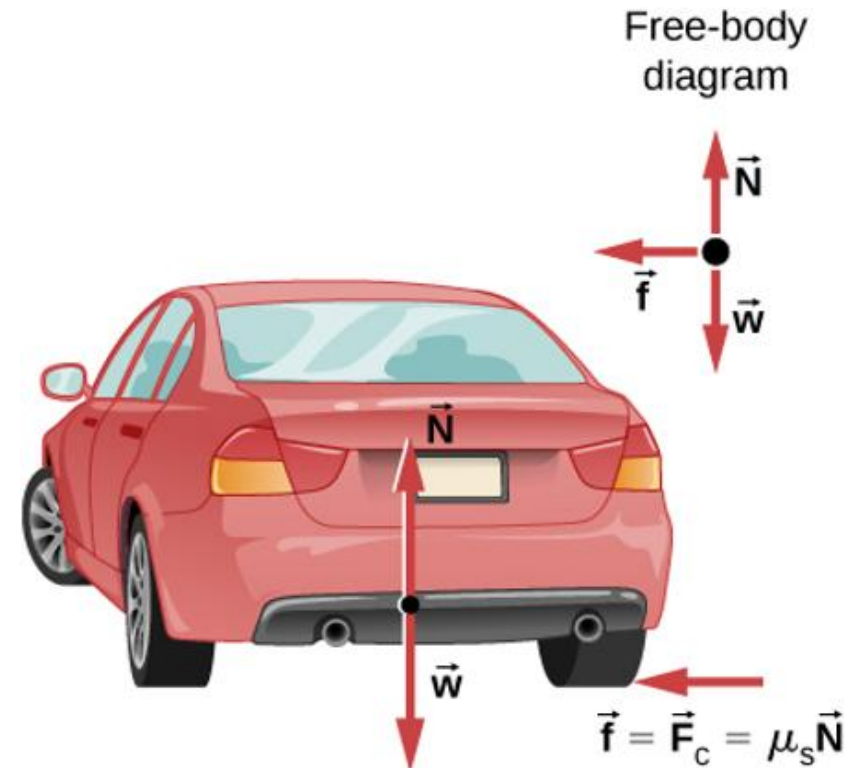
Coefficient of friction for a car on the Flat Curve

A 900 kg car as it negotiates an unbanked curve with a radius of 500 m at a speed of 25.0 m/s. Find the minimum **coefficient of friction** between the tire and the road, static friction being the reason that keeps the car from slipping. [**Answer: $\mu_s = 0.13$**]

$$\mu_s = \frac{v^2}{Rg}$$

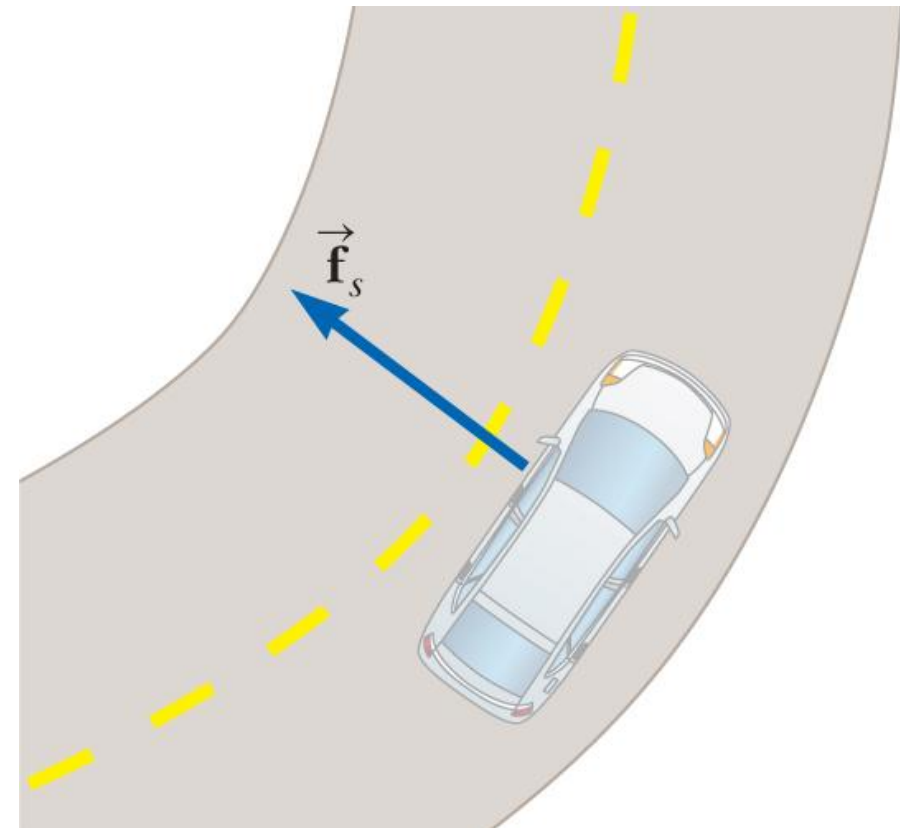
Note: You can find maximum speed to drive a car in the curve, if R and μ_s is provided using the following.

$$v_{\max} = \sqrt{\mu_s Rg}$$



Example 2 (What is the maximum speed of a car?)

(a) A 1,500 kg car moving on a flat, horizontal road negotiates a curve as shown in the overhead view in the figure. If the radius of the curve is 35.0 m and the coefficient of static friction between the tires and dry pavement is 0.523, find the **maximum speed** the car can have and still make the turn successfully.



$$v_{\max} = \sqrt{\mu_s R g}$$

Example 2 (What is the maximum speed of a car?)

(The force that is directed to the center)

$$F_C = ma_c$$

$$f_s = \frac{mv^2}{R}$$

$$f_{s,\max} = \frac{mv_{\max}^2}{R}$$

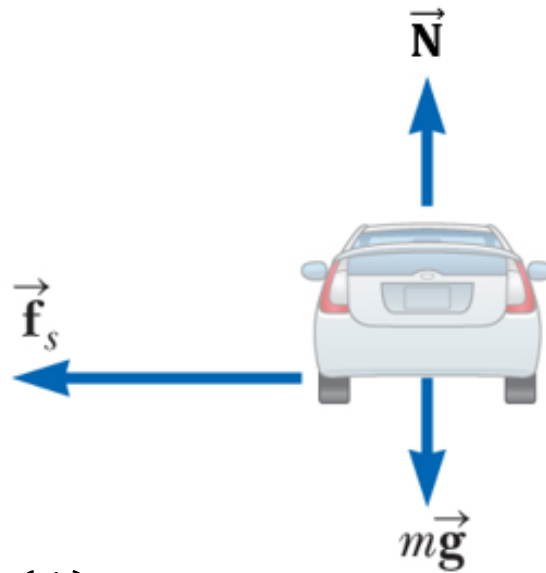
$$\mu_s N = \frac{mv_{\max}^2}{R} \dots \dots (1)$$

(No motion along y-axis)

$$\sum F_y = 0$$

$$N - mg = 0$$

$$N = mg \dots \dots (2)$$



From equation (1) and (2)

$$\mu_s mg = \frac{mv_{\max}^2}{R}$$

$$\mu_s mg R = mv_{\max}^2$$

$$\mu_s g R = v_{\max}^2 \dots \dots (3)$$

$$v_{\max} = \sqrt{\mu_s g R}$$

$$v_{\max} = \sqrt{(0.523)(9.80)(35.0)}$$

$$v_{\max} = 13.4 \text{ m/s}$$

Example 2 (What is the maximum speed of a car?)

(b) Suppose a car travels this curve on a wet day and begins to skid on the curve when its speed reaches only 8.00 m/s. What can we say about the **coefficient of static friction** in this case?

$$\mu_s g r = v_{\max}^2 \dots \dots (3)$$

$$\mu_s = \frac{v_{\max}^2}{g r}$$

$$\mu_s = \frac{(8.00)^2}{(9.80)(35.0)} = 0.187$$

Car on a Banked Curve (ignore friction)

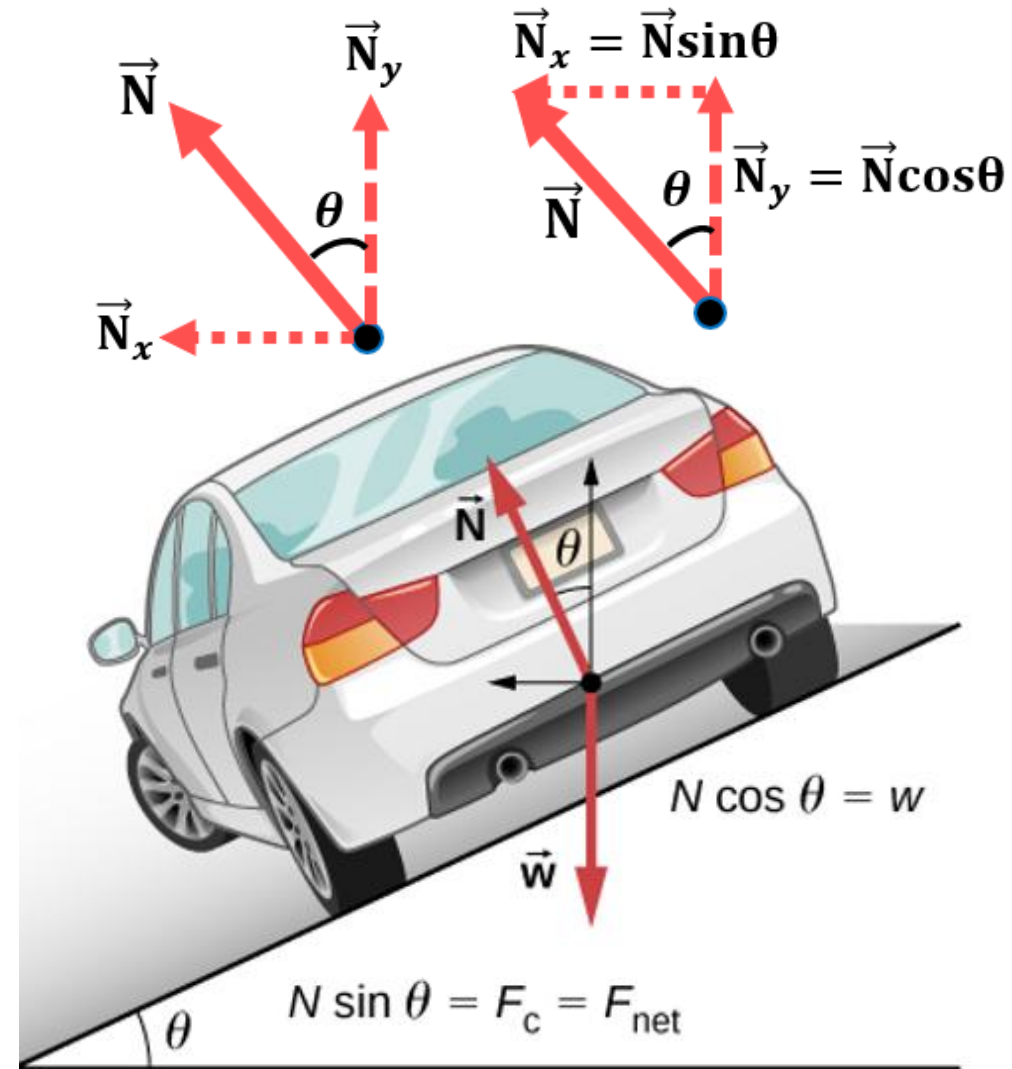
$$\left. \begin{aligned} F_c &= \frac{mv^2}{R} \\ F_c &= N \sin \theta \end{aligned} \right\} N \sin \theta = \frac{mv^2}{R} \dots \dots (1)$$

$$N \cos \theta = mg \dots \dots (2)$$

Dividing equation (1) by equation(2),

$$\frac{\sin \theta}{\cos \theta} = \frac{mv^2}{mgR}$$

$$\tan \theta = \frac{v^2}{Rg} \dots \dots (3)$$



Example 3 (Banked Road)

(The force that is directed to the center)

$$F_C = ma_c$$

$$N \sin \theta = \frac{mv^2}{r} \dots \dots (1)$$

(No motion along y-axis)

$$\sum F_y = 0$$

$$N \cos \theta - mg = 0$$

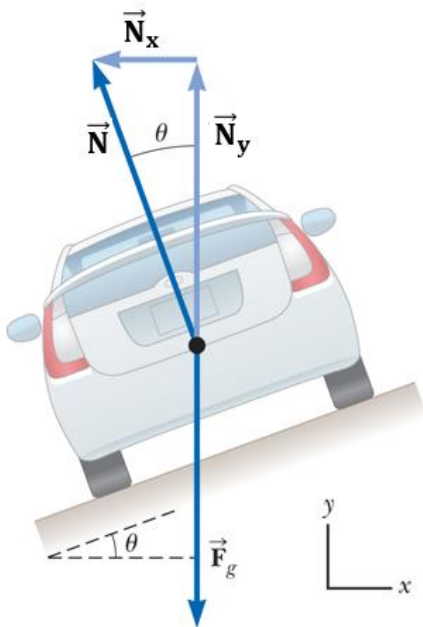
$$\Rightarrow N \cos \theta = mg \dots \dots (2)$$

Dividing equation (1) by (2)

$$\tan \theta = \frac{v^2}{rg}$$

$$\theta = \tan^{-1} \left[\frac{(13.4 \text{ m/s})^2}{(35.0 \text{ m})(9.80 \text{ m/s}^2)} \right]$$

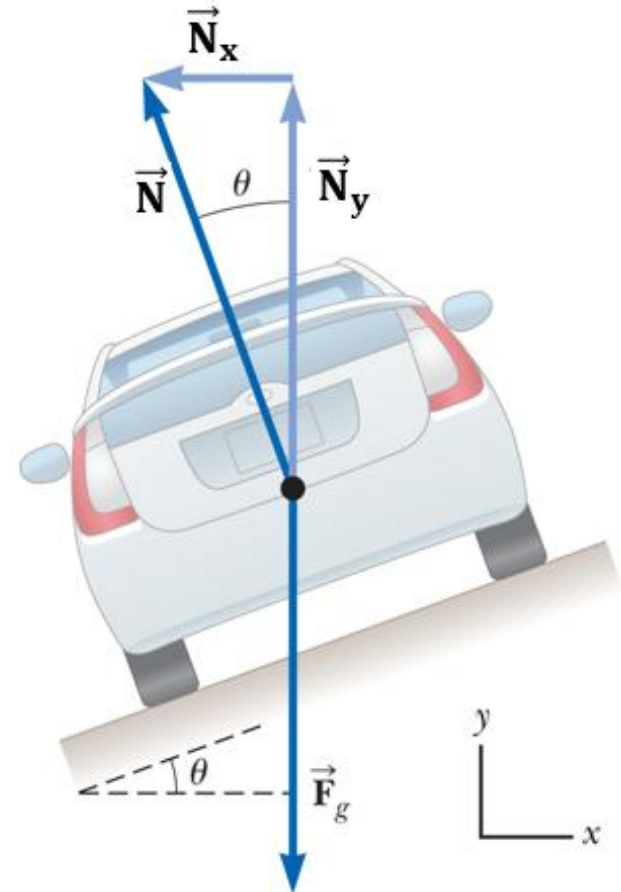
$$\theta = \boxed{27.6^\circ}$$



Example 3 (Banked Road)

You are a civil engineer who has been given the assignment to design a curved roadway in such a way that a car will not have to rely on friction to round the curve without skidding. In other words, a car moving at the designated speed can negotiate the curve when the road is covered with ice. Such road is usually *banked*, which means that the roadway is tilted toward the inside of the curve.

Suppose the designated speed for the road is to be 13.4 m/s (30.0 mi/h) and the radius of the curve is 35.0 m. You need to determine the **angle** at which the roadway on the curve should be banked.



Example#3 (Banked Road)

(The force that is directed to the center)

$$F_C = ma_c$$

$$N \sin \theta = \frac{mv^2}{r} \dots \dots (1)$$

Dividing equation (1) by (2)

$$\tan \theta = \frac{v^2}{rg}$$

$$\theta = \tan^{-1} \left[\frac{(13.4 \text{ m/s})^2}{(35.0 \text{ m})(9.80 \text{ m/s}^2)} \right]$$

$$\theta = \boxed{27.6^\circ}$$

(No motion along y-axis)

$$\sum F_y = 0$$

$$N \cos \theta - mg = 0$$

$$\Rightarrow N \cos \theta = mg \dots \dots (2)$$

Car on a Banked Curve with Friction (Recitation only!)

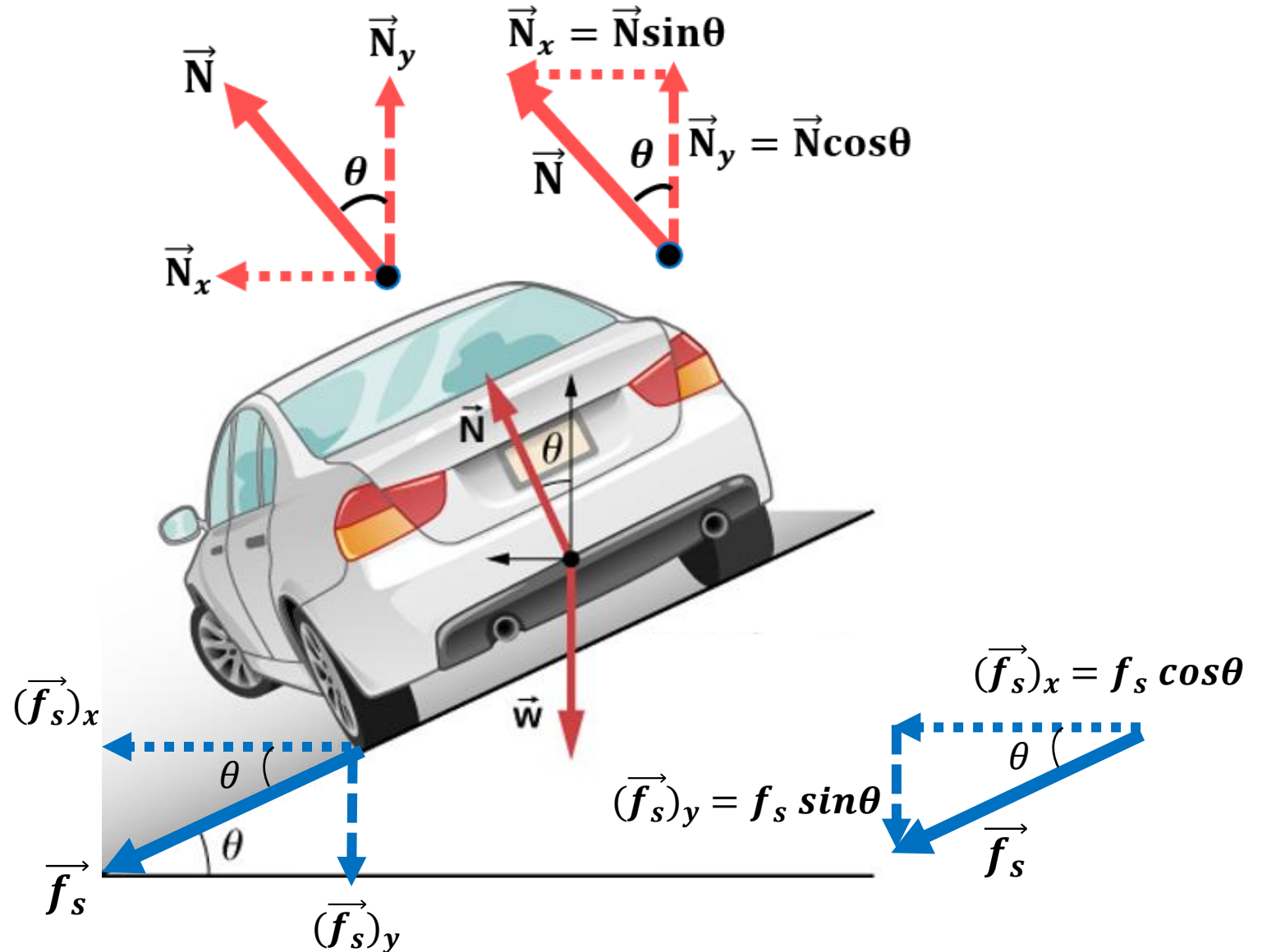
$$F_C = ma_c$$

$$N \sin \theta + f_s \cos \theta = \frac{mv^2}{r}$$

$$N \sin \theta + \mu_s N \cos \theta = \frac{mv^2}{r}$$

For Lecture (ignore friction)

$$N \sin \theta + \mu_s N \cos \theta = \frac{mv^2}{r}$$



Banked Curve - NASCAR



Drag Force and terminal velocity



Drag Force

Definition: a force that resists (R) an object going through the air

Variable: R or F_D

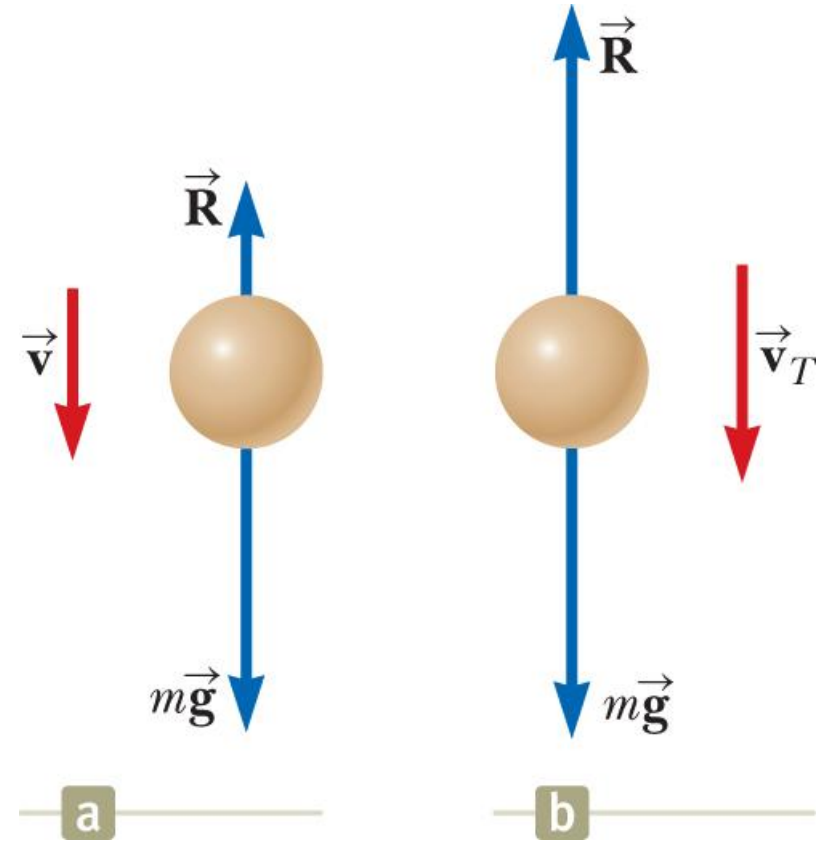
Unit: Newtons (N)

Equation: $F_D = \frac{1}{2}C\rho Av^2$

- C : coefficient of drag
- ρ : density of fluid
- A : area of object facing fluid
- v : velocity of object

(**terminal velocity**)

Direction: Opposite direction of motion



Examples of Fluids and drag coefficient

Object	c
Airfoil	0.05
Toyota Camry	0.28
Ford Focus	0.32
Honda Civic	0.36
Ferrari Testarossa	0.37
Dodge Ram pickup	0.43
Sphere	0.45
Hummer H2 SUV	0.64
Skydiver (feet first)	0.70
Bicycle	0.90
Skydiver (horizontal)	1.0
Circular flat plate	1.12

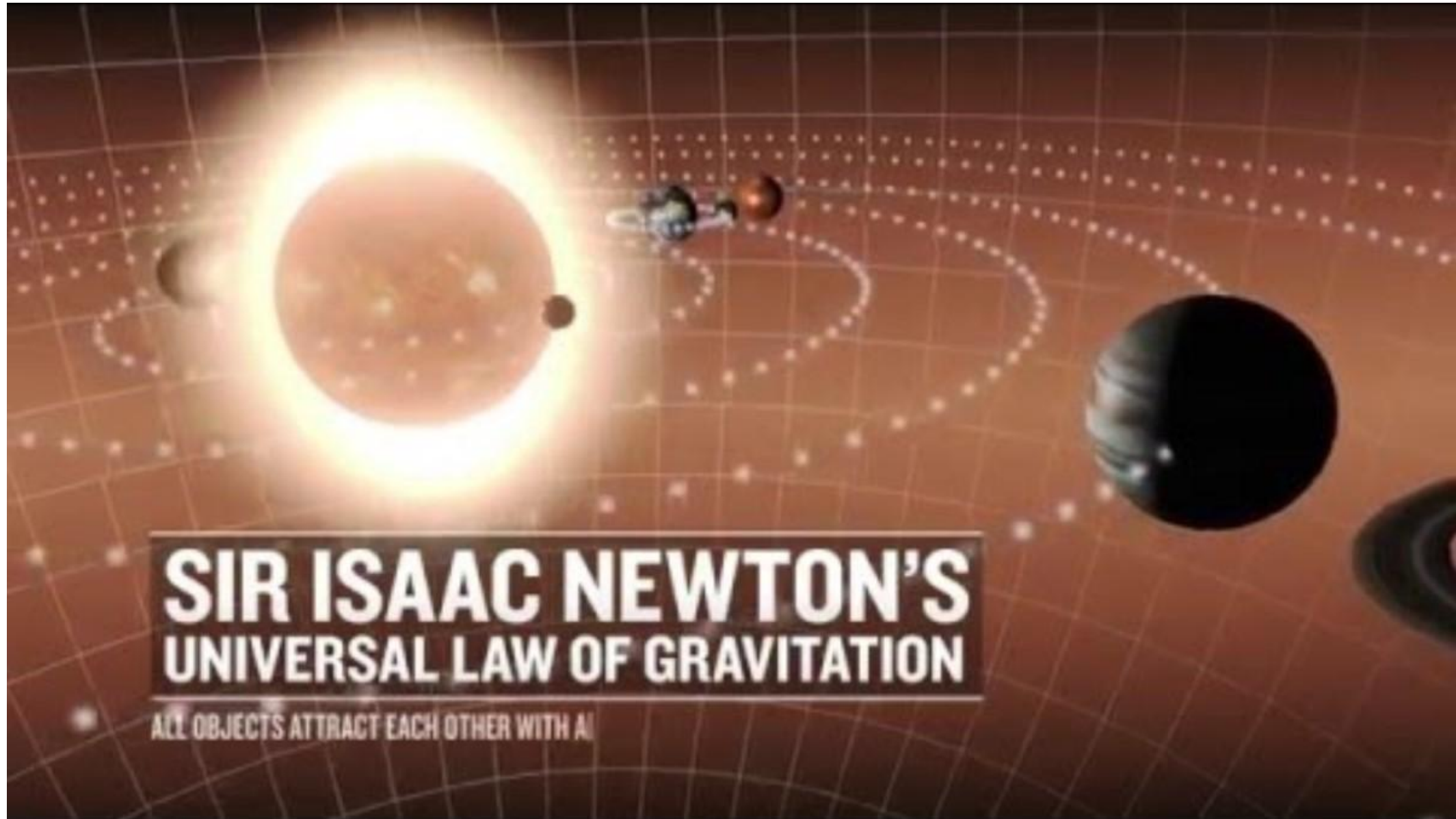
Chapter 13: Outline

13.1 Newton's Law of Universal Gravitation

13.2 Gravitation Near Earth's Surface

13.4 Satellite Orbit

Newton's Law of Universal Gravitation



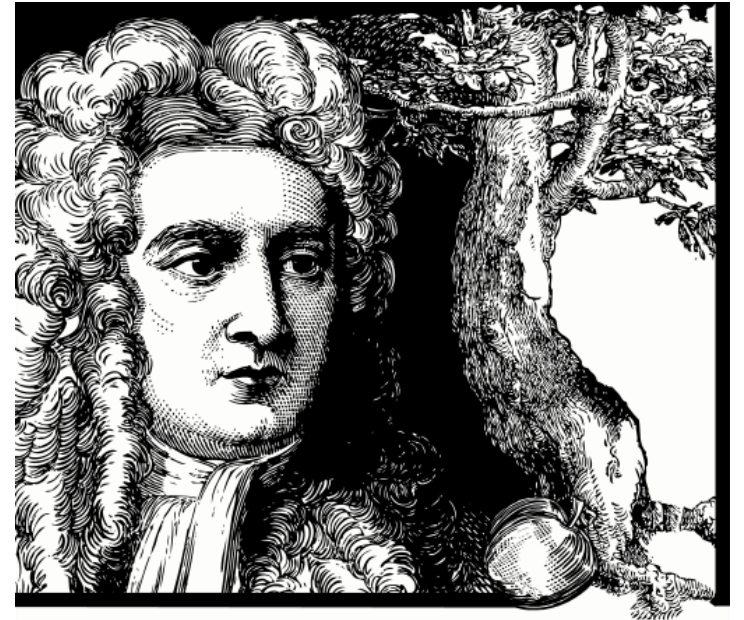
Newton's Law of Universal Gravitation

Every particle in the Universe attracts every other particle with a force that is directly proportional to the product of their masses and inversely proportional to the square of the distance between them.

$$F_g = G \frac{m_1 m_2}{r^2}$$

$$G = 6.674 \times 10^{-11} \text{ N} \cdot \text{m}^2 / \text{kg}^2$$

(**universal gravitation constant**)

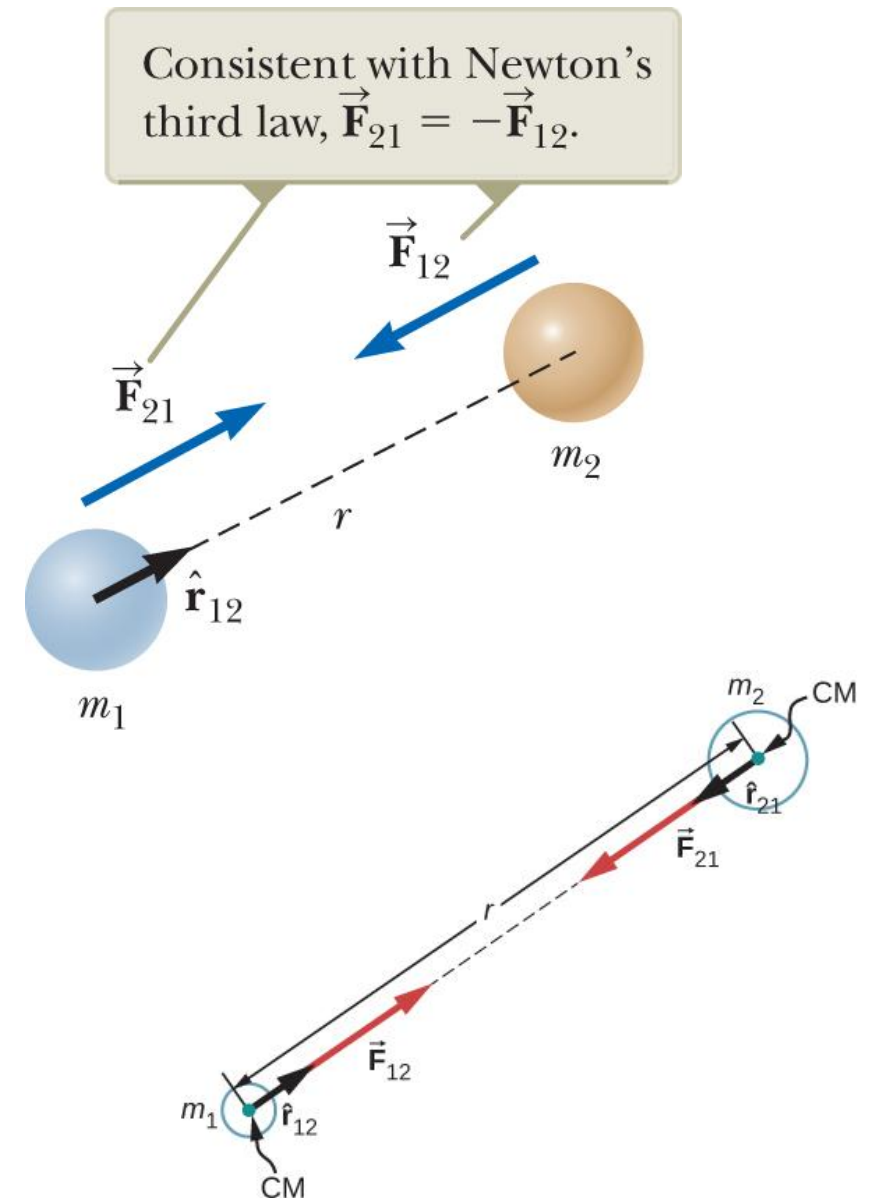


Newton's Law of Universal Gravitation

- Magnitude of force varies as inverse square of separation of particles:

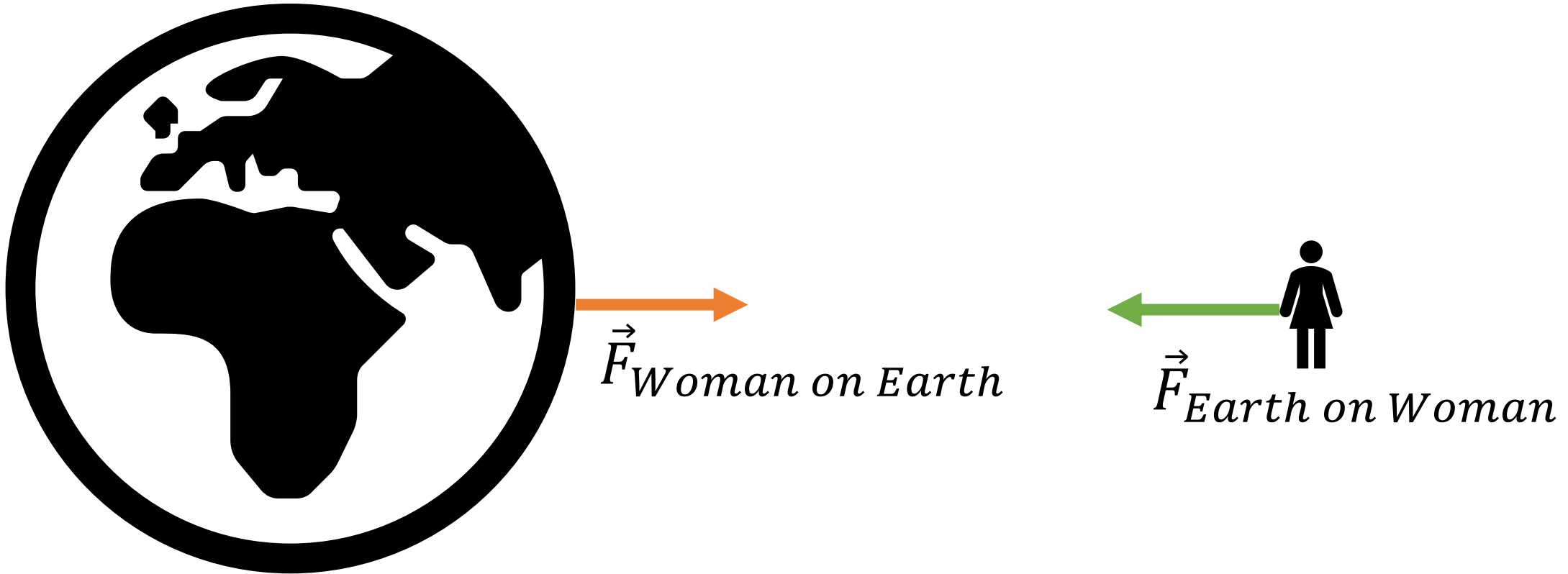
$$F_g = G \frac{m_1 m_2}{r^2}$$

$$\vec{\mathbf{F}}_{12} = -G \frac{m_1 m_2}{r^2} \hat{\mathbf{r}}_{12}$$



Comparing Forces

Newton's 3rd Law says that for every force, there is an equal and opposite so they must be the same!



13.2 Gravitation near the surface of Earth

- Force exerted by Earth on particle of mass “m” near Earth’s surface:

$$F_g = G \frac{M_E m}{R_E^2} \dots (1)$$

- From Newton’s 2nd Law **Weight** to be the following:

$$F_g(\text{or } W) = mg \dots (2)$$

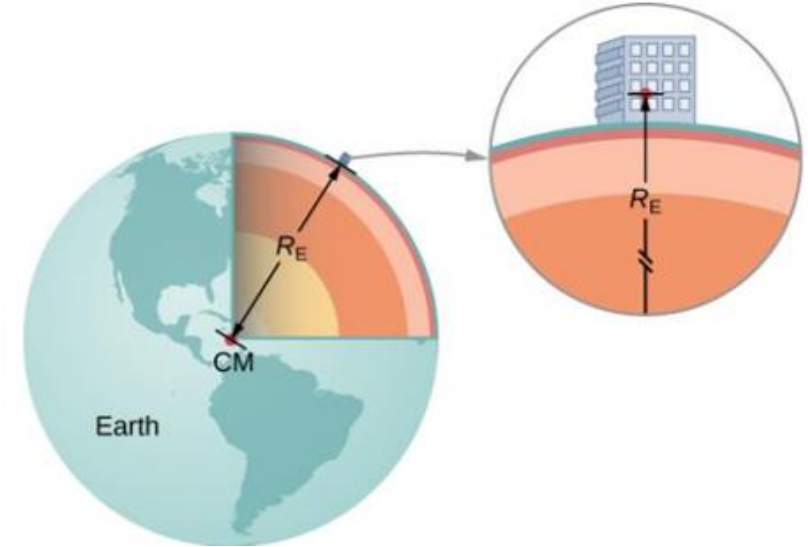
- From (1) and (2)

$$g = G \frac{M_E}{R_E^2}$$

$$R_E = 6.37 \times 10^6 \text{ m}$$

$$M_E = 5.97 \times 10^{24} \text{ kg}$$

$$G = 6.674 \times 10^{-11} \text{ N.m}^2/\text{kg}^2$$



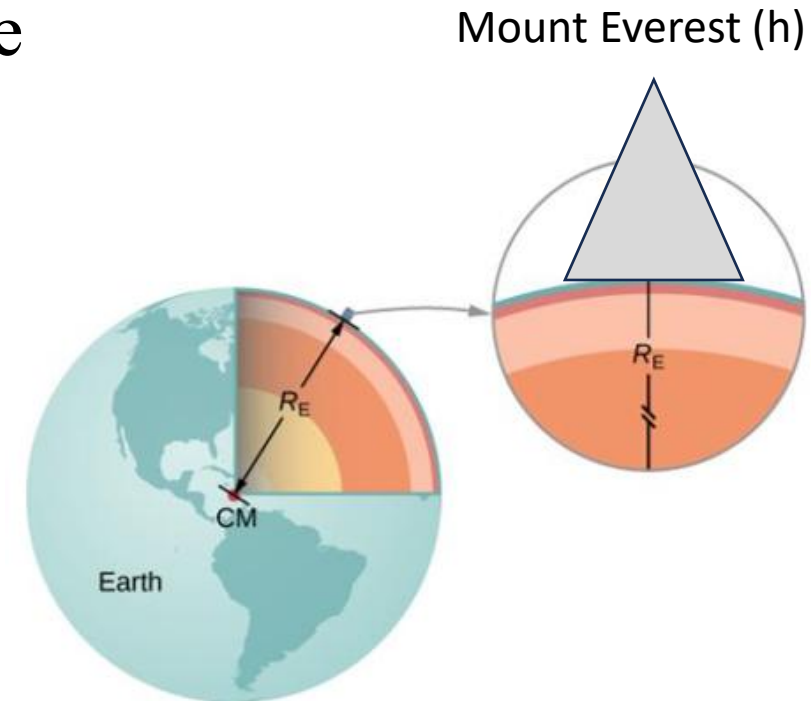
- We can take the distance between the **center of mass of Earth** and an object on its surface to be the radius of Earth, provided that its size is much less than the radius of Earth.

13.2 Gravitation near the surface of Earth

- Object of mass m located distance h above Earth's surface ($r = R_E + h$):

$$F_g = G \frac{M_E m}{r^2} = G \frac{M_E m}{(R_E + h)^2}$$

$$g = \frac{GM_E}{r^2} = \frac{GM_E}{(R_E + h)^2}$$



$$R_E = 6.37 \times 10^6 \text{ m} \quad M_E = 5.97 \times 10^{24} \text{ kg} \quad G = 6.674 \times 10^{-11} \text{ N.m}^2/\text{kg}^2$$

13.2 Gravitation near the surface of Earth

- Object of mass m located distance h above Earth's surface ($r = R_E + h$):

$$F_g = G \frac{M_E m}{r^2} = G \frac{M_E m}{(R_E + h)^2}$$

$$g = \frac{GM_E}{r^2} = \frac{GM_E}{(R_E + h)^2}$$

Free-Fall
Acceleration g at Various
Altitudes Above
the Earth's Surface

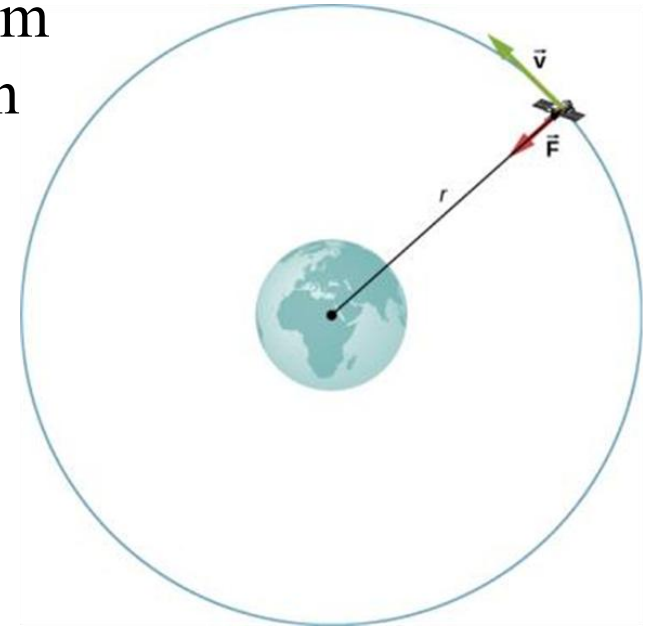
Altitude h (km)	g (m/s ²)
0	9.80
1 000	7.33
2 000	5.68
3 000	4.53
4 000	3.70
5 000	3.08
6 000	2.60
7 000	2.23
8 000	1.93
9 000	1.69
10 000	1.49
50 000	0.13
∞	0

Example #13.4 (Gravity above the Earth's surface)

What is the value of **acceleration due to gravity** “g” 400 km above Earth’s surface, where the International Space Station (ISS) is in orbit?

$$r = R_E + h$$

$$g = \frac{GM_E}{r^2} = \frac{GM_E}{(R_E + h)^2}$$



$$\begin{aligned} h &= 400 \text{ km} \\ &= 400 \times 1000 \text{ m} \\ &= 4.00 \times 10^5 \text{ m} \end{aligned}$$

$$g = \frac{(6.674 \times 10^{-11} \text{ N} \cdot \text{m}^2/\text{kg}^2)(5.97 \times 10^{24} \text{ kg})}{(6.37 \times 10^6 + 400 \times 10^3)^2 \text{ m}}$$

$$g = 8.67 \text{ m/s}^2$$

$$R_E = 6.37 \times 10^6 \text{ m} \quad M_E = 5.97 \times 10^{24} \text{ kg} \quad G = 6.674 \times 10^{-11} \text{ N} \cdot \text{m}^2/\text{kg}^2$$

Example #13.4 (Gravity above the Earth's Surface)

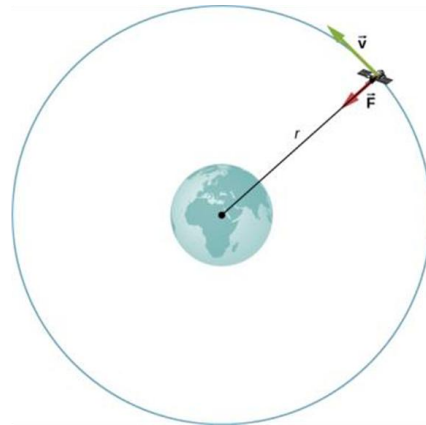
What is the value of “g” 4.00×10^2 km above the Earth's surface, where the International Space Station (ISS) is in orbit? ($M_E = 5.96 \times 10^{24} \text{ kg}$)

$$r = (R_E + h)$$

$$g = \frac{GM_E}{r^2} = \frac{GM_E}{(R_E + h)^2}$$

$$g_s = \frac{(6.67 \times 10^{-11} \text{ N}\cdot\text{m}^2/\text{kg}^2)(5.96 \times 10^{24} \text{ kg})}{(6.37 \times 10^6 + 400 \times 10^3 \text{ m})^2}$$

$$g_s = 8.67 \text{ m/s}^2$$



Method II:

$$g = \frac{GM_E}{r^2} = \frac{GM_E}{(R_E + h)^2}$$

$$g_s = \frac{GM_E}{(R_E + h)^2} \times \frac{(R_E)^2}{(R_E)^2}$$

$$= \frac{GM_E}{(R_E)^2} \times \frac{(R_E)^2}{(R_E + h)^2}$$

$$= 9.80 \text{ m/s}^2 \times \frac{(R_E)^2}{(R_E + h)^2}$$

$$g_s = 8.67 \text{ m/s}^2$$

ABCD card

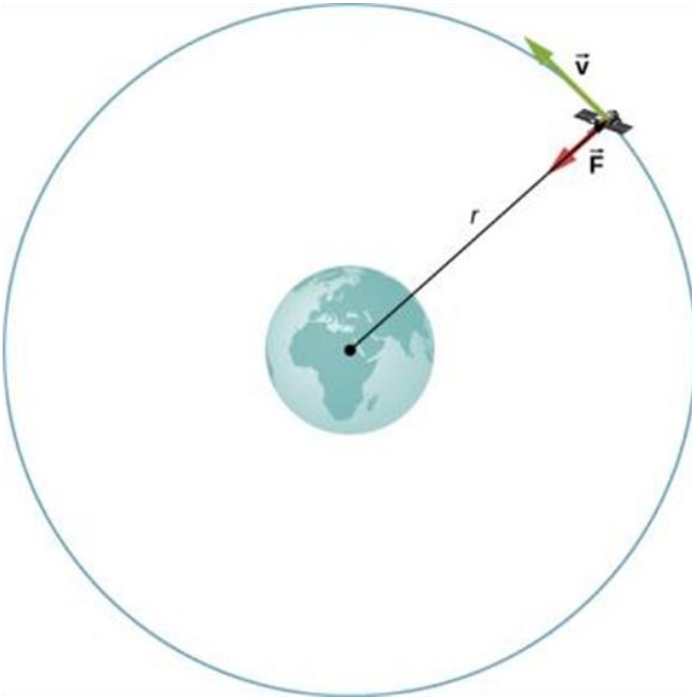
An astronaut orbits the earth in a space capsule whose height above the earth is equal to the earth's radius. How does the **weight** of the astronaut in the capsule compare to her weight on the Earth?

$$h = R_E$$
$$r = (R_E + h) = 2R_E$$

- A. Her weight is equal to zero newtons.
- B. Her weight is approximately equal to one-fourth her weight on Earth
- C. Her weight is approximately equal to one-half of her weight on Earth.
- D. Her weight is approximately equal to one-third of her weight on Earth.
- E. Her weight is approximately equal to one-sixteenth her weight on Earth

$$W = F_g = \frac{G M_E m}{(R_E)^2}$$
$$W_{\text{new}} = \frac{G M_E m}{(R_E + R_E)^2} = \frac{G M_E m}{(2R_E)^2}$$

13.4 Satellite Orbits



- A **satellite of mass m** orbiting at **distance r** from the center of Earth. The gravitational force supplies the centripetal acceleration.
- It has centripetal acceleration directed toward the center of Earth. Earth's gravity is the only force acting, so Newton's second law gives:

$$\frac{GmM_E}{r^2} = ma_c = \frac{mv_{orbit}^2}{r}$$

- We solve for the speed of the orbit, noting that m cancels, and we can get the **speed** and the **period** of the circular orbit:

For circular Motion:

$$v = \frac{2\pi r}{T}$$

$$v_{orbit} = \sqrt{\frac{GM_E}{r}}$$

$$T = 2\pi \sqrt{\frac{r^3}{GM_E}}$$

13.4 Satellite Orbits

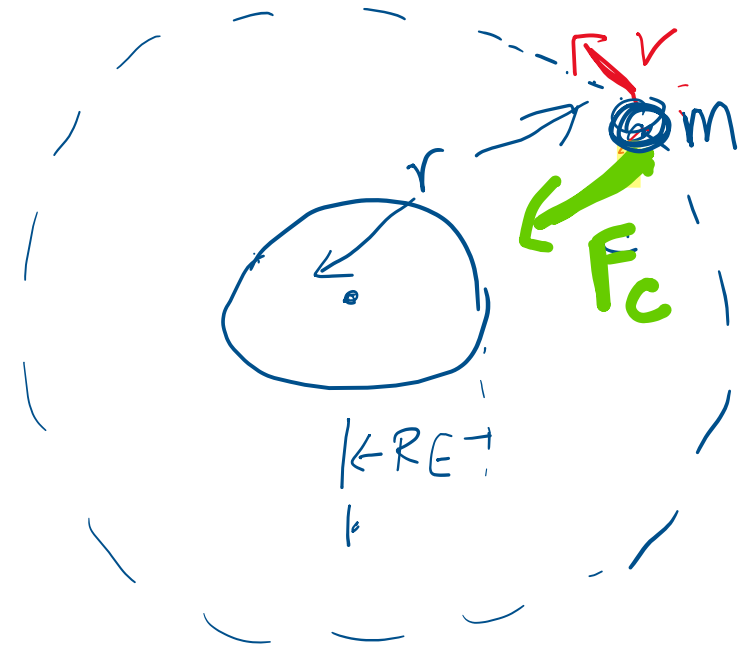
$F_c = m a_c$
Force towards the center

$$F_g = m \frac{v^2}{r}$$

$$\frac{G M_E m}{r^2} = \frac{m v^2}{r}$$

$$\frac{G M_E}{r} = v^2 \quad (*)$$

$$\Rightarrow \boxed{V_{orbit} = \sqrt{\frac{G M_E}{r}}}$$



velocity of the
satellite at the
distance ' r '

13.4 Satellite Orbits

From $\odot$

$$\frac{GM_E}{r} = v^2 \quad \text{---} \odot$$

$$\frac{GM_E}{r} = \left(\frac{2\pi r}{T} \right)^2$$

~~r~~ $\times$ $\frac{GM_E}{\cancel{r}} = \frac{4\pi^2 r^2}{T^2} \times r$

$$T^2 = \frac{4\pi^2 r^3}{GM_E}$$

$$v = \frac{2\pi r}{T}$$

v = velocity

r = radius

T = period

Period of the
satellite at the
distance ' r '

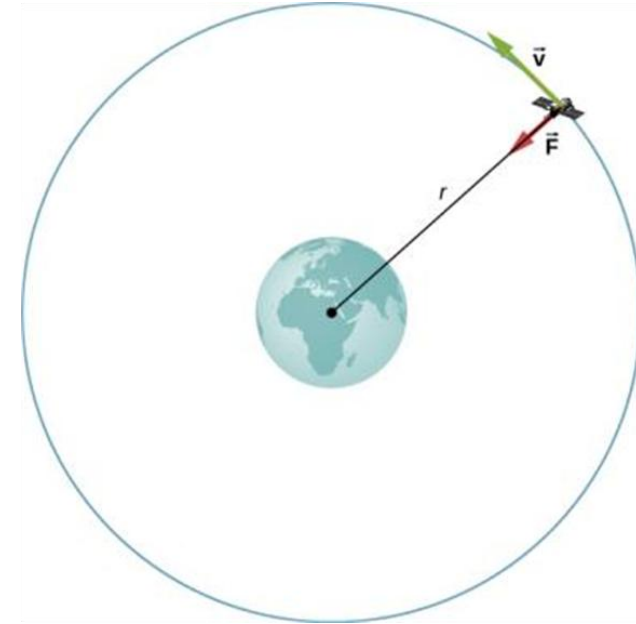
$$T = 2\pi \sqrt{\frac{r^3}{GM_E}}$$

Example #13.9 (International Space Station)

Determine the orbital speed and period of ISS which is orbiting 4.00×10^2 km from the Earth's surface.

$$r = R_E + h$$

$$r = R_E + 4.00 \times 10^2 \text{ km}$$



$$v_{orbit} = \sqrt{\frac{GM_E}{r}} = \sqrt{\frac{(6.674 \times 10^{-11} \text{ N} \cdot \text{m}^2/\text{kg}^2)(5.97 \times 10^{24} \text{ kg})}{(6.36 \times 10^6 \text{ m} + 4.00 \times 10^5 \text{ m})}} = 7.67 \times 10^3 \text{ m/s}$$

$$T = 2\pi \sqrt{\frac{r^3}{GM_E}} = 2\pi \sqrt{\frac{(6.36 \times 10^6 \text{ m} + 4.00 \times 10^5 \text{ m})^3}{(6.674 \times 10^{-11} \text{ N} \cdot \text{m}^2/\text{kg}^2)(5.97 \times 10^{24} \text{ kg})}} = 5.55 \times 10^3 \text{ s}$$

$$R_E = 6.37 \times 10^6 \text{ m} \quad M_E = 5.97 \times 10^{24} \text{ kg} \quad G = 6.674 \times 10^{-11} \text{ N} \cdot \text{m}^2/\text{kg}^2$$

Example #3 (Planet Nutron)

The period of a satellite circling planet Nutron is observed to be 84 s when it is in a circular orbit with a radius of 8.0×10^6 m. What is the mass of planet Nutron? $G = 6.67 \times 10^{-11} \text{ N}\cdot\text{m}^2/\text{kg}^2$.

[Answer: $4.3 \times 10^{28} \text{ kg}$]

Hint: Gravitational force = Centripetal force

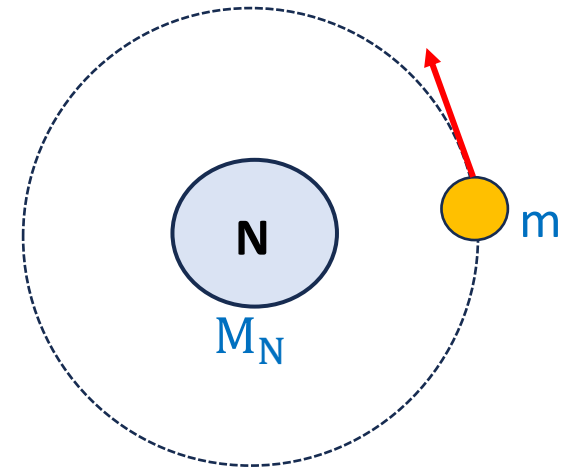
$$\frac{G M_N m}{r^2} = \frac{m v^2}{r}$$

$$\begin{aligned} T &= 84 \text{ s} \\ r &= 8.0 \times 10^6 \text{ m} \\ M_N &= ? \end{aligned}$$

$$G M_N = r v^2$$

$$M_N = \frac{r}{G} v^2$$

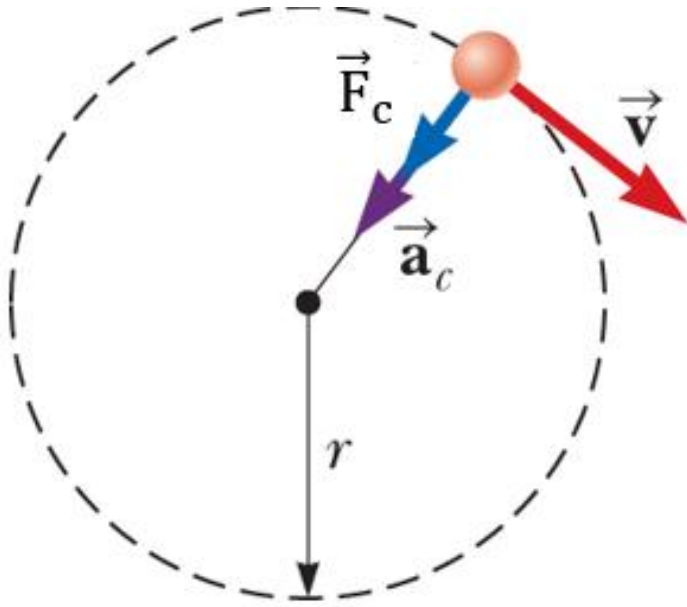
$$= \frac{r}{G} \left(\frac{4\pi^2 r^2}{T^2} \right) = \frac{4\pi^2 r^3}{G \cdot T^2}$$



For circular Motion:

$$v = \frac{2\pi r}{T}$$

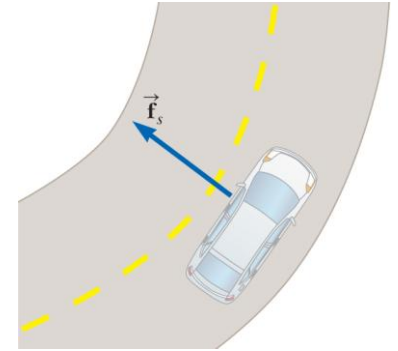
Summary: Circular Motion



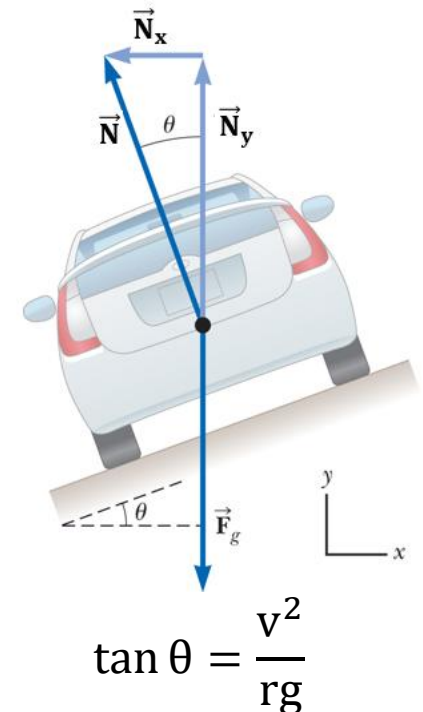
Centripetal Force:

$$F_C = ma_c = m \frac{v^2}{r}$$

- Identify a circular motion (the center of the circle, direction of the acceleration and velocity).
- What provides the centripetal force for a circular motion?
- Figure out the required condition for a circular motion, including tension force in a rope, static friction force (coefficient), and banked angle of a curve?



Flat curve:
static friction



Summary: Circular Motion

- Universal gravitation:

$$F_g = G \frac{m_1 m_2}{r^2}$$

- Free-fall acceleration:

$$mg = G \frac{M_E m}{R_E^2} \Rightarrow g = G \frac{M_E}{R_E^2} \quad g = \frac{GM_E}{r^2} = \frac{GM_E}{(R_E + h)^2}$$

- Satellite Orbits:

$$\frac{GmM_E}{r^2} = ma_c = \frac{mv_{orbit}^2}{r}$$

$$v_{orbit} = \sqrt{\frac{GM_E}{r}}$$

$$T = 2\pi \sqrt{\frac{r^3}{GM_E}}$$